

Worksheet 6A: Statistics and Probability

AIML Foundations Mathematics

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***What This Worksheet Is About** Statistics and probability are the mathematical foundation of machine learning. Every time an AI makes a prediction, classifies an image, or recommends a video, it's using these concepts. This worksheet connects the basics you've learned to real applications — from training neural networks to analyzing scientific data.*

Part A: Measures of Central Tendency — Mean, Median, Mode (18 problems)

The Formulas

Mean (Average): $\bar{x} = \frac{\sum x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$

Median: The middle value when data is sorted (or average of two middle values if n is even)

Mode: The most frequent value

Calculate the mean, median, and mode:

1. Test scores: 78, 82, 85, 85, 88, 90, 92
2. Response times (ms): 120, 135, 140, 145, 145, 150, 180
3. Daily website visitors: 1200, 1350, 1400, 1450, 1500, 1550, 8500 - (a) Calculate all three measures. - (b) Which measure is most affected by the outlier (8500)? - (c) Which measure best represents a "typical" day? **AI Connection: Training Data Quality**
4. A dataset of house prices (in €1000s) contains: 250, 275, 290, 310, 325, 340, 2500. The last value is a mansion that got mixed in with regular houses. - (a) Calculate the mean with and without the outlier. - (b) Calculate the median with and without the outlier. - (c) Why might a data scientist remove outliers before training a model?
5. An image classification dataset has these label counts:

Category	Count
Cat	5000
Dog	4800
Bird	1200
Fish	500

- (a) What is the mean number of images per category? - (b) What is the median? - (c) What is the mode? - (d) This is called an "imbalanced dataset." Why might this be a problem for training? **Science Connection: Experimental Measurements**
- 6. A physicist measures the speed of light in 5 trials (in units of 10^8 m/s): 2.998, 2.995, 3.001, 2.997, 2.999 - (a) Calculate the mean. - (b) The accepted value is 2.998. How close is the experimental mean?
- 7. A chemist measures the pH of a solution 7 times: 7.02, 6.98, 7.01, 6.99, 7.00, 7.01, 7.00 - (a) Find mean, median, and mode. - (b) Are they all similar? What does this suggest about the measurement consistency? **Game Analytics**
- 8. Player scores in a mobile game level: 1250, 1480, 1520, 1550, 1600, 1650, 1700, 1800, 1850, 2400 - (a) Find the mean and median. - (b) If the game designer wants to set the "3-star threshold" at a score that about half of players achieve, should they use the mean or median?
- 9. Daily active users for an app over two weeks: Week 1: 10000, 12000, 15000, 14000, 13000, 8000, 7000 Week 2: 11000, 13000, 16000, 15000, 14000, 9000, 8000 - (a) Calculate the mean for each week. - (b) Is the app growing? By what percentage? **Weighted Mean:**
- 10. A student's grade is calculated as:
 - Homework: 25% weight, score 85
 - Midterm: 30% weight, score 78
 - Final: 45% weight, score 88

Calculate the weighted mean (final grade):

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$$

- 11. In a neural network, outputs from three layers are combined with weights:
 - Layer 1: output 0.7, weight 0.5
 - Layer 2: output 0.3, weight 0.3
 - Layer 3: output 0.8, weight 0.2

Calculate the weighted average output.

Part B: Measures of Spread — Range, Variance, Standard Deviation (16 problems)

The Formulas

Range: $R = x_{max} - x_{min}$

Variance: $\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$

Standard Deviation: $\sigma = \sqrt{\sigma^2}$

Note: For samples, divide by $(n - 1)$ instead of n — this is called Bessel's correction.

Calculate the range, variance, and standard deviation:

12. Data: 4, 6, 8, 10, 12 - (a) Find the mean. - (b) Find $(x_i - \bar{x})$ for each value. - (c) Find $(x_i - \bar{x})^2$ for each value. - (d) Find the variance (divide by $n = 5$). - (e) Find the standard deviation.
13. Data: 7, 7, 8, 8, 8, 9, 9
14. Data: 2, 5, 8, 11, 14 **AI Connection: Model Consistency**
15. Two machine learning models are tested 5 times each on accuracy (%): Model A: 85, 87, 84, 86, 88 Model B: 70, 95, 75, 90, 80 - (a) Calculate the mean accuracy for each. - (b) Calculate the standard deviation for each. - (c) Which model is more consistent? - (d) In production, would you prefer a consistent model or one with higher peaks?
16. A chatbot's response times (seconds) over 6 queries: Version 1.0: 1.2, 1.3, 1.1, 1.4, 1.2, 1.3 Version 2.0: 0.8, 2.1, 0.7, 1.8, 0.9, 1.7 - (a) Which version has lower mean response time? - (b) Which version is more consistent (lower standard deviation)? - (c) Users hate unpredictable wait times. Which version would you deploy? **Science Connection: Measurement Precision**
17. Two instruments measure the same voltage (in volts): Instrument A: 5.02, 4.98, 5.01, 4.99, 5.00 Instrument B: 5.10, 4.90, 5.15, 4.85, 5.00 - (a) Both have the same mean. Verify this. - (b) Calculate the standard deviation for each. - (c) Which instrument is more precise?
18. In manufacturing, parts must be within tolerance. A machine produces bolts with target diameter 10.00 mm. A sample of 5 bolts measures: 10.02, 9.98, 10.01, 9.99, 10.00 - (a) Find the mean and standard deviation. - (b) If tolerance is ± 0.05 mm (so 9.95 to 10.05 is acceptable), are all bolts within tolerance? - (c) If the standard deviation were 0.03 mm instead, what percentage of bolts would you expect to be outside tolerance? (Assume normal distribution where 68% are within 1 SD of mean) **Understanding Standard Deviation:**
19. For a normal distribution:
 - 68% of data falls within 1 standard deviation of the mean
 - 95% falls within 2 standard deviations
 - 99.7% falls within 3 standard deviations

IQ scores have mean 100 and standard deviation 15. - (a) What range contains about 68% of people? - (b) What range contains about 95% of people? - (c) What percentage of people have IQ above 130?
20. Human height (adult males) has mean ≈ 175 cm and SD ≈ 7 cm. - (a) What range contains the middle 68%? - (b) How tall would someone need to be to be in the top 2.5%? - (c) A basketball team requires players at least 190 cm tall. Approximately what percentage of men qualify?

Part C: Basic Probability (20 problems)

The Fundamentals

$$P(\text{event}) = \frac{\text{Number of favorable outcomes}}{\text{Total possible outcomes}}$$

Key Rules: - $0 \leq P(A) \leq 1$ - $P(\text{certain event}) = 1$ - $P(\text{impossible event}) = 0$ - $P(\text{not } A) = 1 - P(A)$

21. A standard deck has 52 cards ($4 \text{ suits} \times 13 \text{ values}$). Find:

- (a) $P(\text{drawing a heart})$
- (b) $P(\text{drawing an ace})$
- (c) $P(\text{drawing the ace of hearts})$
- (d) $P(\text{not drawing a face card})$ (face cards = J, Q, K)

22. A bag contains 5 red, 3 blue, and 2 green marbles. Find:

- (a) $P(\text{red})$
- (b) $P(\text{not red})$
- (c) $P(\text{red or blue})$

AI Connection: Classification Probabilities

23. A spam filter classifies 1000 emails:

- 800 are correctly identified (700 real emails marked "not spam", 100 spam marked "spam")
- 200 are incorrectly identified (150 real emails marked "spam", 50 spam marked "not spam")

- (a) What is $P(\text{correct classification})$? - (b) What is $P(\text{false positive})$ — real email marked as spam? - (c) What is $P(\text{false negative})$ — spam marked as real? - (d) Which error is worse for the user?

24. An image classifier sees 500 images:

- 200 cats (correctly identifies 180)
- 200 dogs (correctly identifies 170)
- 100 birds (correctly identifies 80)

- (a) What is $P(\text{correct} \mid \text{cat})$ — probability of correct given it's a cat? - (b) What is $P(\text{correct} \mid \text{dog})$? - (c) What is $P(\text{correct} \mid \text{bird})$? - (d) What is the overall accuracy?

Science Connection: Medical Testing

25. A disease affects 1% of the population. A test for the disease:

- Correctly identifies 95% of sick people (sensitivity)

- Correctly identifies 90% of healthy people (specificity)

In a population of 10,000 people: - (a) How many are actually sick? Healthy? - (b) Of the sick, how many test positive? - (c) Of the healthy, how many test positive (false positives)? - (d) If someone tests positive, what's the probability they're actually sick? *This is the famous "base rate fallacy" — the answer may surprise you!* **Game Probability**

26. A loot box in a game contains:

- Common item: 70% chance
- Rare item: 20% chance
- Epic item: 8% chance
- Legendary item: 2% chance

- (a) What is $P(\text{rare or better})$? - (b) If you open 3 boxes, what's the probability of getting no legendary items? - (c) A player opens 50 boxes and gets no legendary items. How "unlucky" is this? Calculate $P(\text{no legendary in 50 boxes}) = 0.98^{50}$

27. In a card game, you need to draw 2 specific cards from a 20-card deck:

- (a) What is $P(\text{first card is correct})$?
- (b) Given the first card was correct, what is $P(\text{second card is correct})$?
- (c) What is $P(\text{both cards correct})$?

Addition Rule (OR)

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

For mutually exclusive events: $P(A \text{ or } B) = P(A) + P(B)$

28. Rolling a die:

- (a) $P(\text{even or greater than 4})$
- (b) Are "even" and "greater than 4" mutually exclusive?

29. Drawing from a deck:

- (a) $P(\text{heart or face card})$
- (b) Why do we subtract $P(\text{heart and face card})$?

Multiplication Rule (AND)

Independent events: $P(A \text{ and } B) = P(A) \times P(B)$ **Dependent events:** $P(A \text{ and } B) = P(A) \times P(B|A)$

30. Flipping a coin and rolling a die:

- (a) $P(\text{heads and 6})$
- (b) $P(\text{heads and even})$

31. Drawing two cards without replacement:

- (a) $P(\text{both aces})$
- (b) $P(\text{both hearts})$

AI Connection: Independent Events in ML

32. A neural network has 3 layers, each with 95% chance of working correctly (independently). - (a) What is $P(\text{all layers work})$? - (b) What is $P(\text{at least one layer fails})$? - (c) If the network has 10 layers, each 99% reliable, what is $P(\text{all work})$?

33. A self-driving car uses 4 independent sensors:

- Camera: 99% reliable
- LIDAR: 98% reliable
- Radar: 97% reliable
- Ultrasonic: 95% reliable

- (a) What is $P(\text{all sensors work})$? - (b) The car needs at least one sensor working. What is $P(\text{at least one works})$?

Part D: Counting Principles (16 problems)

Fundamental Counting Principle

If task 1 can be done in m ways and task 2 can be done in n ways, then both tasks can be done in $m \times n$ ways.

34. A password has 4 characters:

- First character: letter (26 options)
- Second: digit (10 options)
- Third: letter (26 options)
- Fourth: digit (10 options)

How many possible passwords are there?

35. A restaurant offers:

- 4 appetizers
- 6 main courses
- 3 desserts

How many different 3-course meals are possible?

Permutations (Order Matters)

$$P(n, r) = \frac{n!}{(n - r)!}$$

Number of ways to arrange r items from n items.

36. How many ways can 5 people line up for a photo?
37. How many 3-letter "words" (any letters, no repeats) can be made from A, B, C, D, E?
38. In a race with 8 runners, how many ways can gold, silver, and bronze be awarded?

Combinations (Order Doesn't Matter)

$$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n - r)!}$$

Number of ways to choose r items from n items.

39. How many ways can you choose 3 students from a class of 10 to form a committee?
40. How many 5-card poker hands are possible from a 52-card deck?
41. A lottery requires choosing 6 numbers from 1-49. How many possible tickets? **AI Connection: Feature Selection**
42. A data scientist has 10 features (variables) and wants to train models using different subsets: - (a) How many ways to choose exactly 3 features? - (b) How many ways to choose exactly 5 features? - (c) How many total subsets of features are there (including empty set)? Hint: Each feature is either in or out = 2^{10}
43. A neural network architect is designing a network with:
 - Choice of 3 activation functions
 - Choice of 4 optimization algorithms
 - Number of hidden layers: 1, 2, 3, or 4
 - Learning rate: 5 options

How many different configurations are possible? **Science Connection: Genetics**

44. DNA has 4 bases: A, T, G, C. A codon is a sequence of 3 bases. - (a) How many different codons are possible? - (b) There are 20 amino acids. Why does this work out nicely?
45. In a genetics experiment, a researcher needs to test all pairwise combinations of 8 genes.
 - (a) How many pairs must be tested?
 - (b) If each test takes 2 hours, how long will the full experiment take?

Pascal's Triangle Connection

46. Row 5 of Pascal's Triangle is: 1, 5, 10, 10, 5, 1 These are $\binom{5}{0}, \binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4}, \binom{5}{5}$
Verify: $\binom{5}{2} = 10$ using the formula.
47. The sum of row n of Pascal's Triangle is 2^n . - (a) Verify for row 5: $1+5+10+10+5+1 = ?$ - (b) This represents all possible subsets of a 5-element set. Why?

Part E: Putting It Together — ML Applications (10 problems)

Accuracy, Precision, Recall

48. A model classifying tumors as "malignant" or "benign" produces:

	Predicted Malignant	Predicted Benign
Actually Malignant	85 (True Positive)	15 (False Negative)
Actually Benign	20 (False Positive)	880 (True Negative)

Calculate: - (a) **Accuracy** = $(TP + TN) / \text{Total}$ - (b) **Precision** = $TP / (TP + FP)$ — "Of predicted positives, how many are correct?" - (c) **Recall** = $TP / (TP + FN)$ — "Of actual positives, how many did we catch?" - (d) In cancer detection, is it worse to have low precision or low recall?

49. Another model has:

- Accuracy: 95%
- Precision: 70%
- Recall: 99%

What does this tell you about the model's behavior? **Expected Value**

50. Expected value: $E[X] = \sum x_i \cdot P(x_i)$ A game costs €5 to play. You roll a die: - Roll 6: win €20 - Roll 5: win €10 - Roll anything else: win €0 - (a) Calculate the expected winnings. - (b) What is the expected profit (winnings minus cost)? - (c) Is this a fair game?

51. A trading algorithm has:

- 60% chance of €100 profit
- 25% chance of breaking even
- 15% chance of €200 loss

- (a) Calculate the expected value per trade. - (b) Over 100 trades, what is the expected total profit? **Bayes' Theorem Preview**

52. Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

A factory has two machines: - Machine A produces 60% of items, with 2% defect rate - Machine B produces 40% of items, with 5% defect rate An item is selected at random and found to be defective. What is the probability it came from Machine A? - (a) Find $P(\text{defective}) = P(\text{def}|A)P(A) + P(\text{def}|B)P(B)$ - (b) Apply Bayes' Theorem

Scientific Sampling

53. A biologist wants to estimate the number of fish in a lake using capture-recapture:
- Day 1: Capture 100 fish, tag them, release
 - Day 2: Capture 80 fish, find 10 are tagged

Using the proportion: $\frac{\text{tagged in sample}}{\text{sample size}} \approx \frac{\text{total tagged}}{\text{population}}$ Estimate the total fish population.

Cross-Validation

54. In 5-fold cross-validation, a model's accuracy on each fold is: Fold 1: 88%, Fold 2: 85%, Fold 3: 90%, Fold 4: 87%, Fold 5: 85% - (a) What is the mean accuracy? - (b) What is the standard deviation? - (c) Report the result as "mean $\pm$ SD" **A/B**

Testing

55. A website runs an A/B test on a new checkout button:
- Version A (old): 1000 visitors, 50 purchases (5% conversion)
 - Version B (new): 1000 visitors, 65 purchases (6.5% conversion)
- (a) What is the relative improvement of B over A? - (b) Is this difference "real" or could it be random chance? (This requires statistical tests we'll learn later, but intuitively, is 15 extra purchases out of 1000 a big deal?)